

SMARTX: Porewater Ammonium

May 2026 Samples

2026-08-11

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##Run Information

```
cat("Run Information: ") #lets you know what section you're in
```

Run Information:

```
library(here)
```

here() starts at C:/Users/blkleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW

```
#set the run date & user name
```

```
sample_year <- "2026"
```

```
sample_month <- "May"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month
```

```
NH3_files <- c("Raw Data/20260624_SMARTX_Run1_May2026.csv",  
              "Raw Data/20260626_SMARTX_Run2_May2026.csv",  
              "Raw Data/20260807_SMARTX_Run3_May2026.csv",  
              "Raw Data/20260626_2M_SMARTXrr_May2026.csv"  
              )
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
file_path <- here("Porewater QAQC Logs", "GCREW_NH4_QAQC_Log.csv")
```

```
file_path_QAQC <- here("Porewater QAQC Logs", "GCREW_SEAL_Check_QAQC_Log.csv")
```

```
final_path <- "Processed Data/GCREW_SMARTX_Porewater_NH4_202605.csv"
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- "SMARTX_441-80 Not Collected; One PE Check Passes One Fails 20260626"
```

```
#Add SMARTX Prefix if necessary to samples
```

```
Add_Prefix <- "YES"
```

```
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"
```

```
cat(run_notes)
```

SMARTX_441-80 Not Collected; One PE Check Passes One Fails 20260626

##Setup

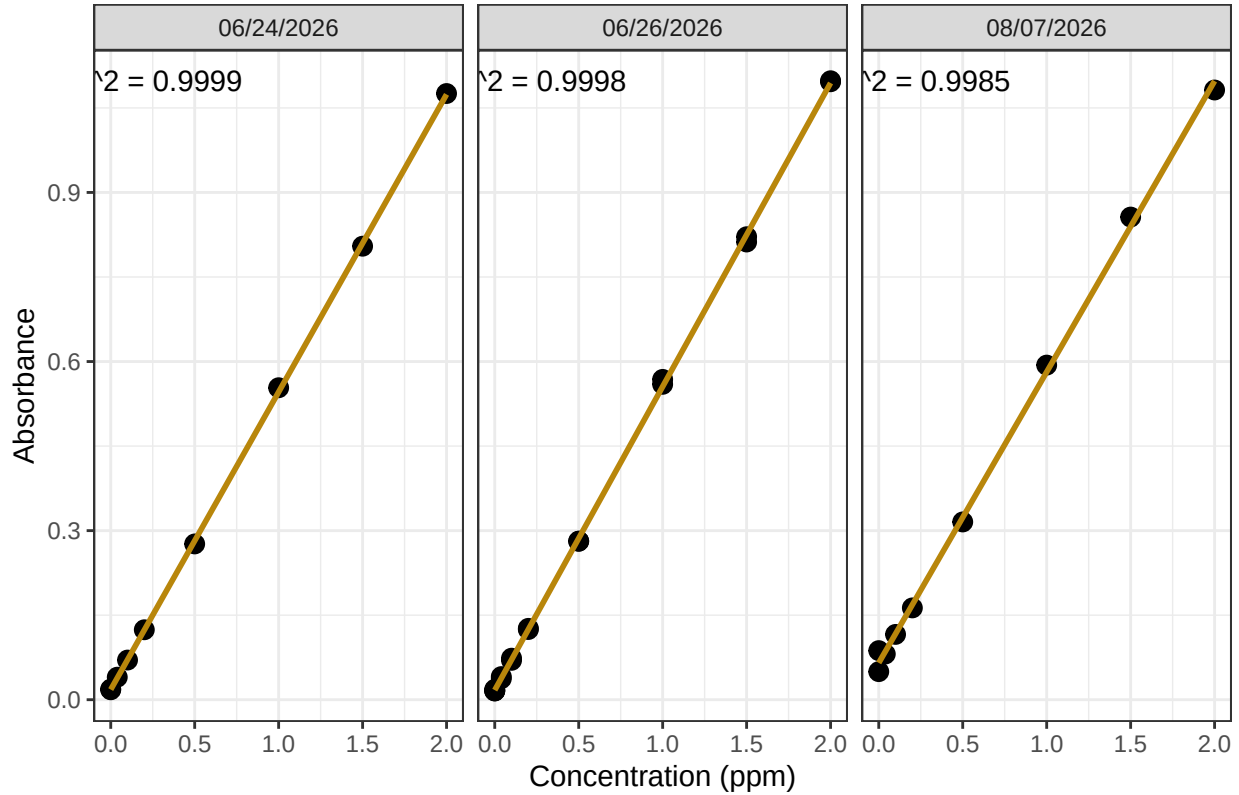
##Read in metadata and create similar sample IDs for matching to samples

```
##Import Data & Clean
```

0.1 Assessing standard Curves

```
## 'geom_smooth()' using formula = 'y ~ x'
```

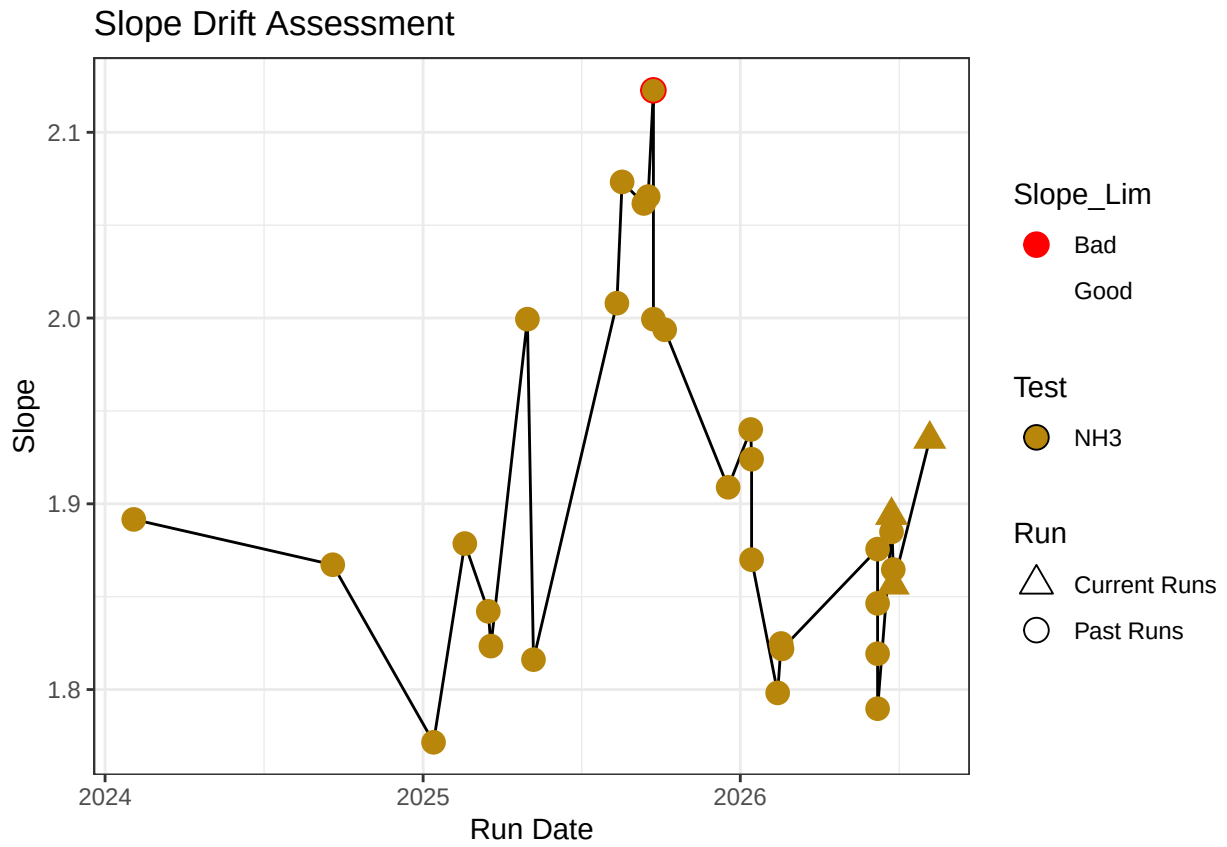
NH3 Standard Curve

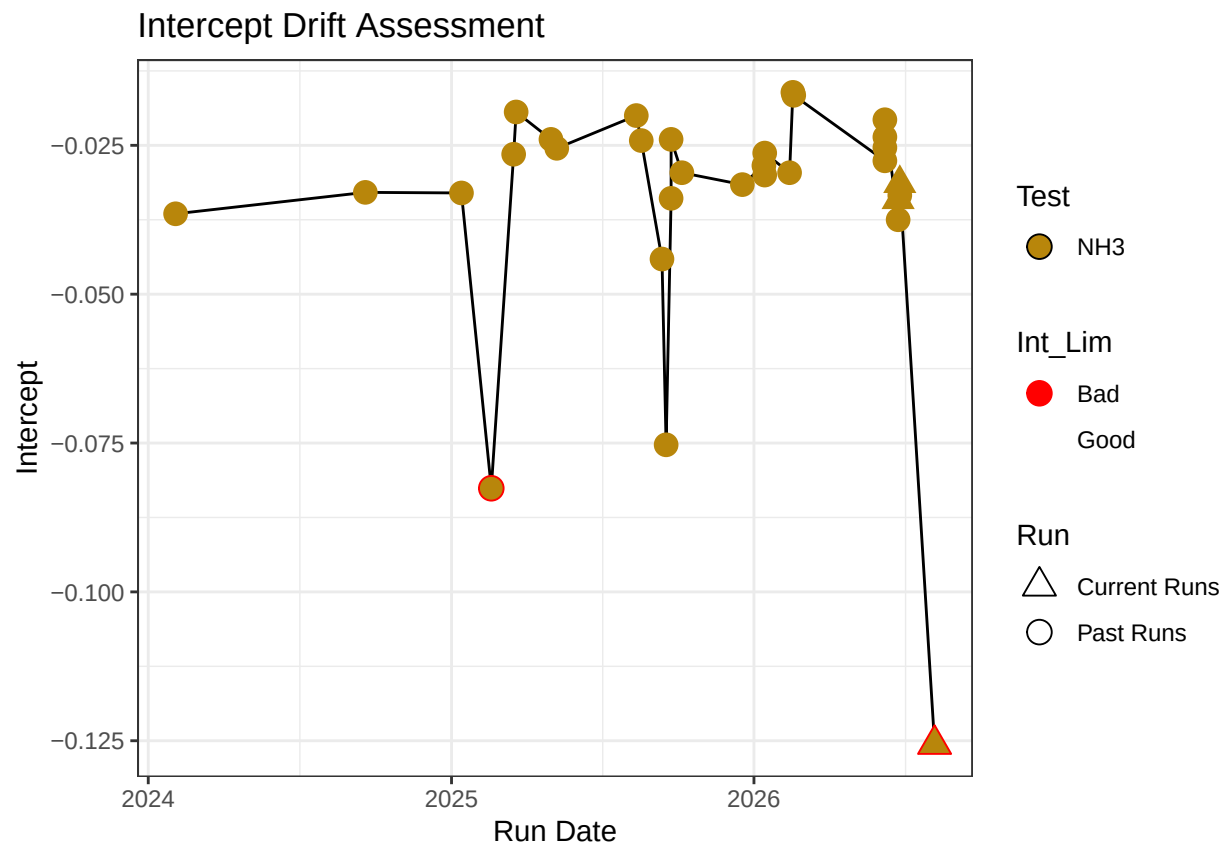


Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	06/24/2026	R2_Pass	0.9998859	1.893843
Ammonia 2	06/26/2026	R2_Pass	0.9998100	1.856283
Ammonia 2	08/07/2026	R2_Pass	0.9985244	1.934832

0.2 Update Standard Log

```
## 'summarise()' has grouped output by 'Test', 'Intercept', 'Slope', 'R_Squared',  
## 'User'. You can override using the '.groups' argument.
```





0.3 Dilution Corrections - ensure the latest dilution is kept

```
## [1] "Reran Samples: c(\"321-20\", \"323-20\", \"331-20\", \"332-20\", \"333-20\", \"633-20\", \"312-20\")"
```

```
## [1] "Diluted Samples: c(\"321-20\", \"323-20\", \"331-20\", \"332-20\", \"333-20\", \"633-20\", \"312-20\")"
```

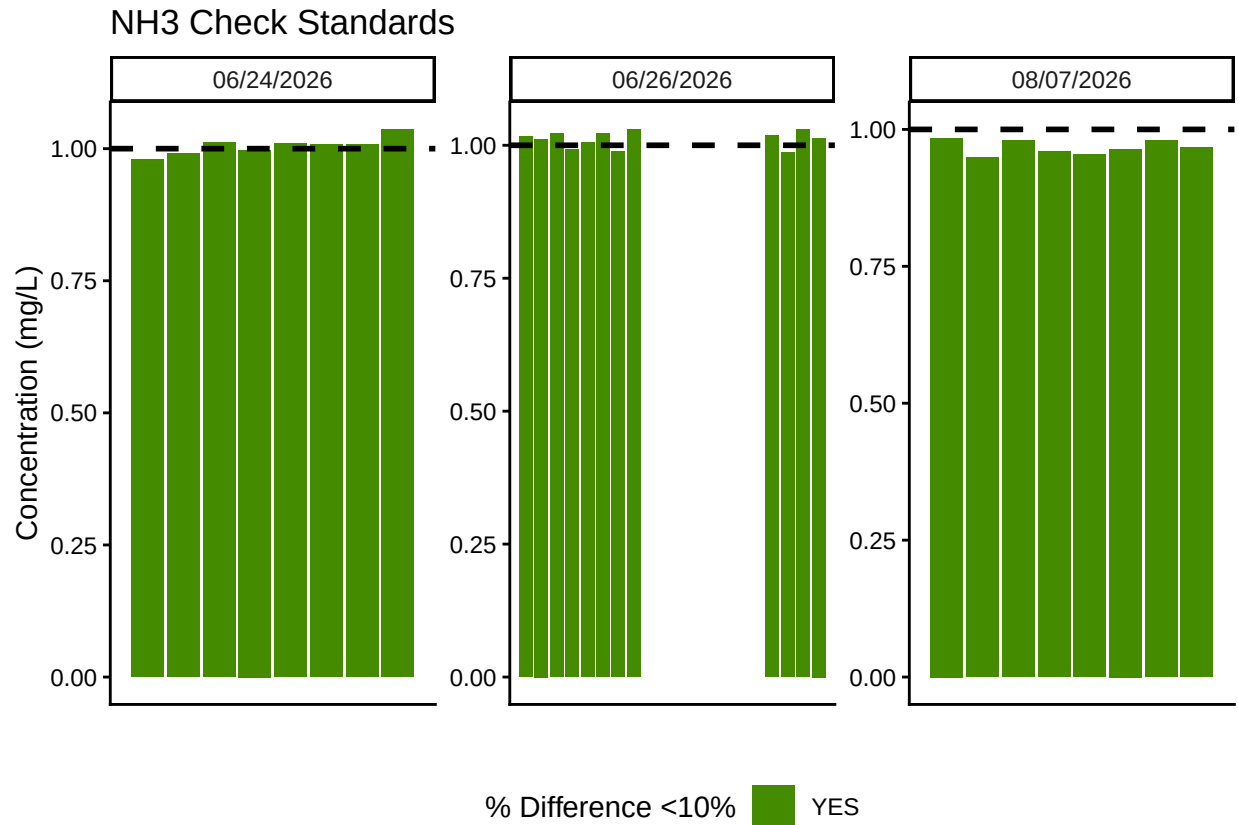
```
## [1] "No Naming Issues Detected"
```

0.4 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	06/24/2026	Performance Check Within 25% - PROCEED	1.086850	1.034
Ammonia 2	06/26/2026	Performance Check NOT Within 25% - REASSESS	1.481548	1.034
Ammonia 2	06/26/2026	Performance Check NOT Within 25% - REASSESS	1.481548	1.034
Ammonia 2	08/07/2026	Performance Check Within 25% - PROCEED	1.096071	1.034

0.5 Analyze the Check Standards

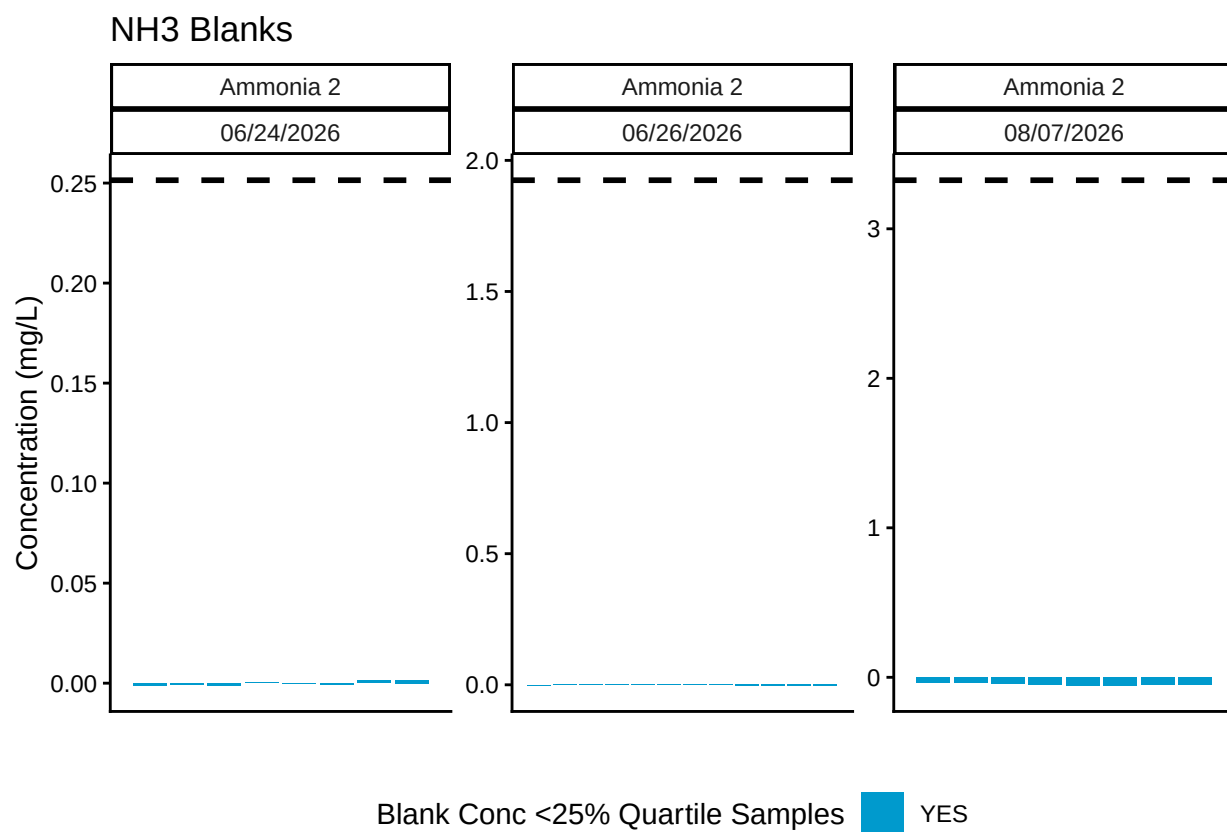
Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	06/24/2026	RSV WITHIN RANGE - PROCEED	0.0242693	0.25
Ammonia 2	06/26/2026	RSV WITHIN RANGE - PROCEED	0.0242693	0.25
Ammonia 2	08/07/2026	RSV WITHIN RANGE - PROCEED	0.0242693	0.25



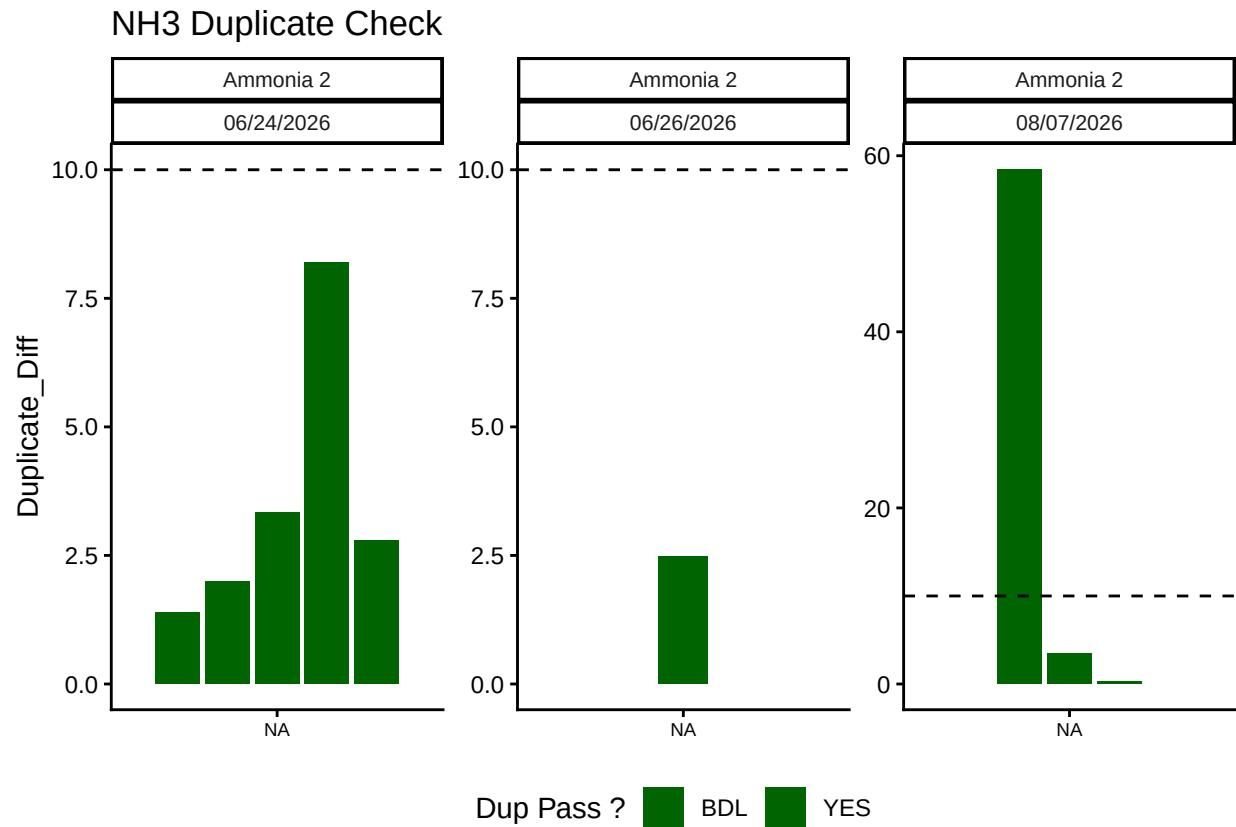
Test	Run_Date	CHK_Flag
Ammonia 2	06/24/2026	>60% of Checks Pass - PROCEED
Ammonia 2	06/26/2026	>60% of Checks Pass - PROCEED
Ammonia 2	08/07/2026	>60% of Checks Pass - PROCEED

0.6 Analyze Blanks

Test	Run_Date	BLK_Pct_Flag	Mean_Blkc_Conc	Quantile_25
Ammonia 2	06/24/2026	>60% of Blanks Pass - PROCEED	-0.0001170	0.2514288
Ammonia 2	06/26/2026	>60% of Blanks Pass - PROCEED	-0.0015411	1.9243220
Ammonia 2	08/07/2026	>60% of Blanks Pass - PROCEED	-0.0484810	3.3251115

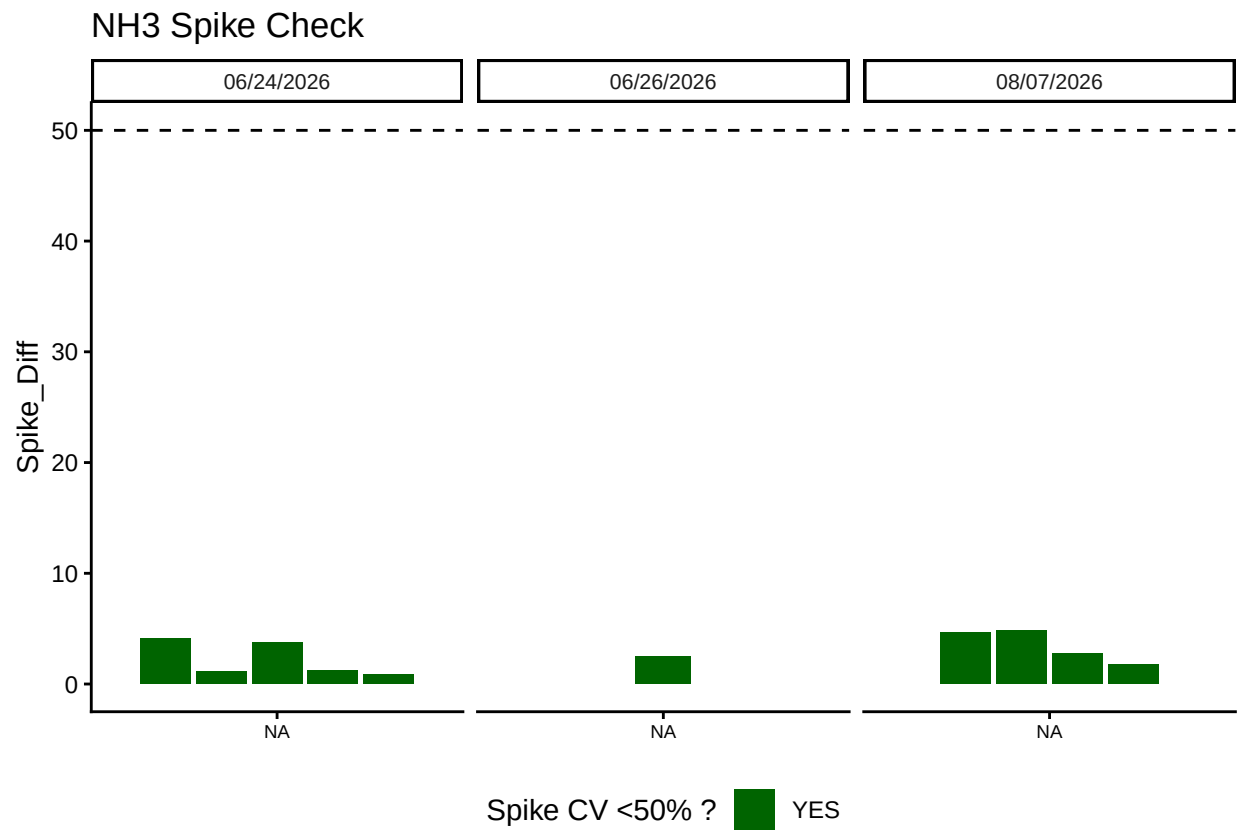


0.7 Analyze Duplicates



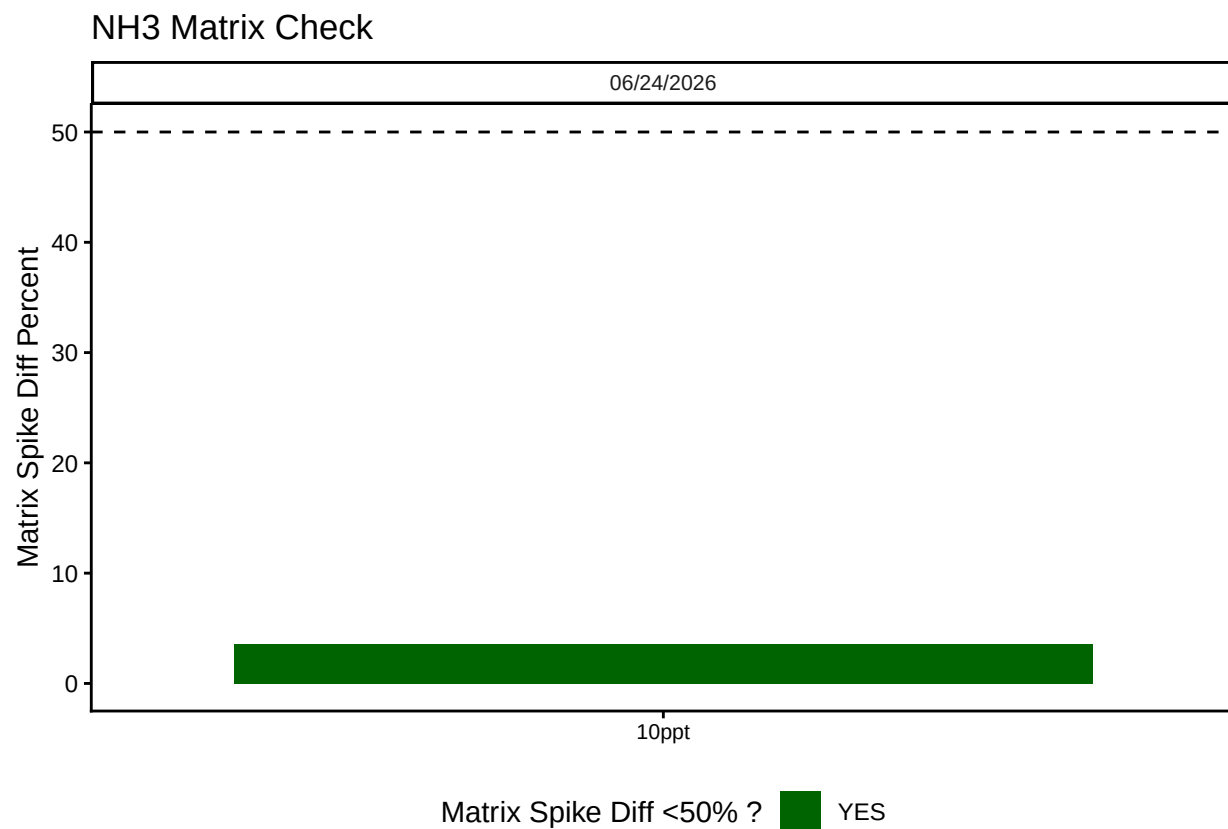
Test	Run_Date	Dup_Flags
Ammonia 2	06/24/2026	>60% of Dups Pass - PROCEED
Ammonia 2	06/26/2026	>60% of Dups Pass - PROCEED
Ammonia 2	08/07/2026	>60% of Dups Pass - PROCEED

0.8 Spikes



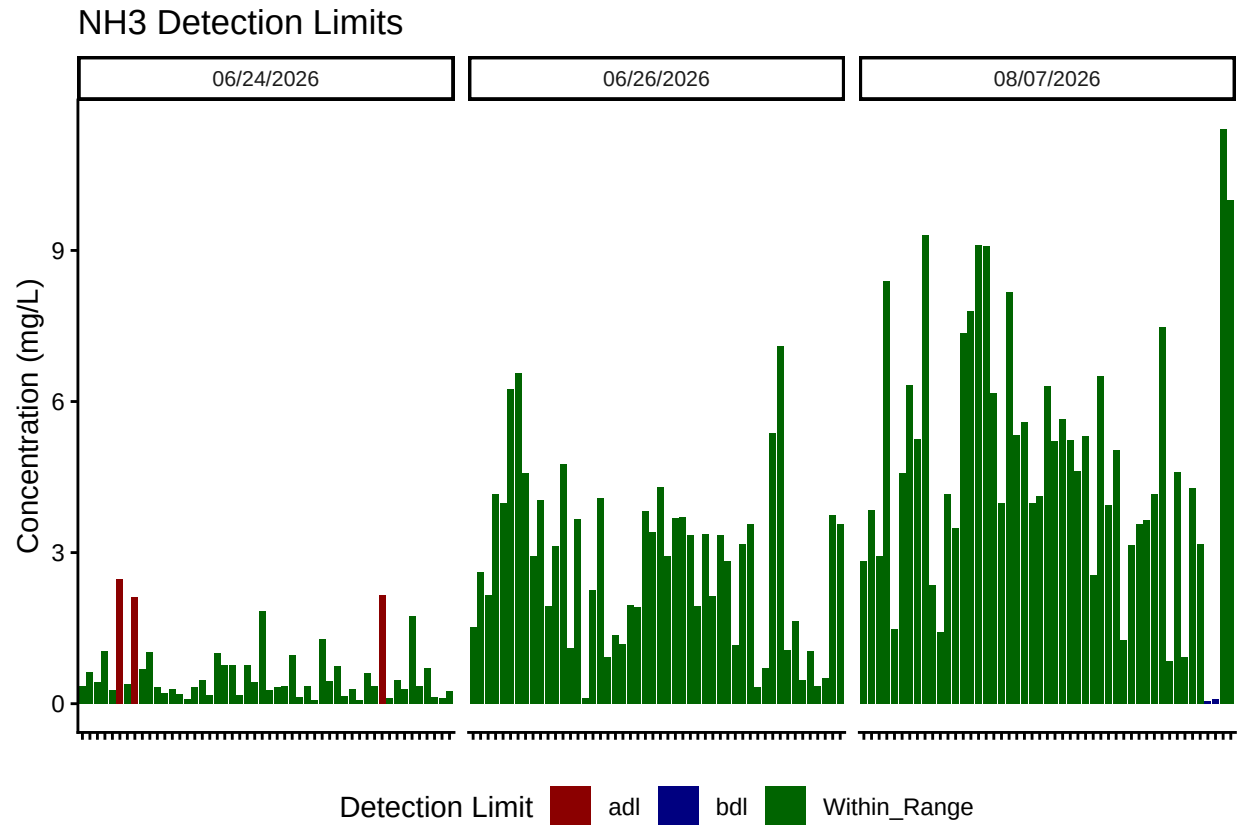
Test	Run_Date	Spike_Flags
Ammonia 2	06/24/2026	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	06/26/2026	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	08/07/2026	>60% of Spikes have a Diff <50% - PROCEED

0.9 Matrix Effects



##Unit Conversion

##Sample Flagging - Within range of standard curve



##Add QAQC Flags

0.10 Create Log for QAQC

```
#Old_Log_QAQC <- tibble(Test = c(), Mean_PE_Conc= c(), Spike_CV= c(), Top_Std_Absorbance= c(), Run_Date= c())
Old_Log_QAQC <- read.csv(file_path_QAQC) %>% mutate(Run_Date = as.Date(as.character(Run_Date), tryFormats = c("%m/%d/%Y", "%m/%d/%y", "%d/%m/%Y", "%d/%m/%y")))

Top_Standards <- Slope_raw %>% group_by(Test, Run_Date) %>% filter(Sample_Name == "Standard 2.0000") %>% summarise(Mean_PE_Conc = mean(Mean_PE_Conc))

PE_Check <- NH3_peChk %>% group_by(Test, Run_Date) %>% dplyr::select(Test, Run_Date, Mean_Chk_Conc) %>% summarise(Mean_Chk_Conc = mean(Mean_Chk_Conc))

Spikes_Log <- Spike_plot_df %>% group_by(Test, Run_Date) %>% dplyr::select(Test, Run_Date, Spike_Diff) %>% summarise(Spike_Diff = mean(Spike_Diff), Run_Date = Run_Date, Spike_Percent_Diff = mean(Spike_Percent_Diff)) %>% ungroup()

Joined_QAQC <- left_join(Top_Standards, PE_Check, by = join_by(Test, Run_Date)) %>%
  left_join(Spikes_Log, by = join_by(Test, Run_Date)) %>% mutate(Run = "Current Runs") %>% mutate(Run_Date = Run_Date)
  group_by(dplyr::select(., -Run)) %>%
  summarize(Run = Run[max(1, which(Run != "Past Runs"))][1]) %>%
  ungroup()
```

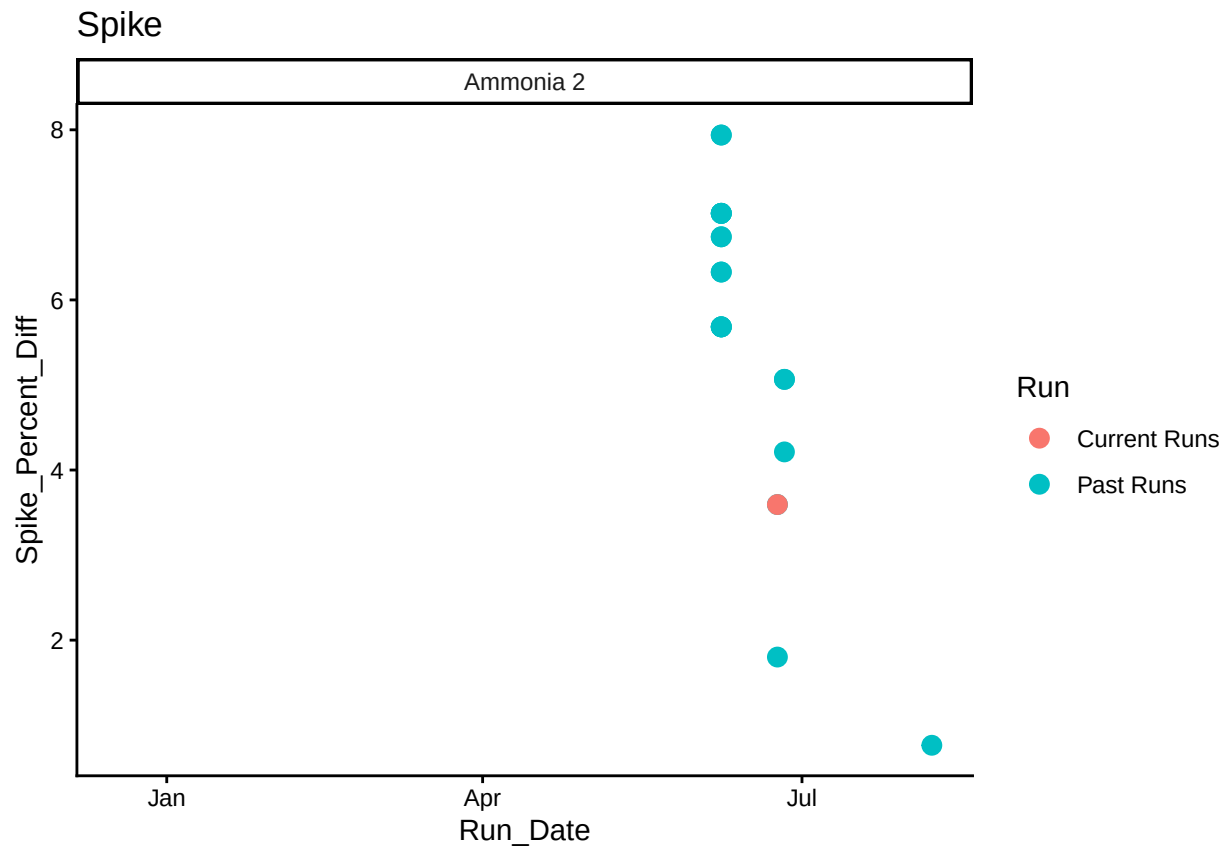
```
## Warning in left_join(Top_Standards, PE_Check, by = join_by(Test, Run_Date)): Detected an unexpected relationship between 'x' and 'y'.
## i Row 2 of 'x' matches multiple rows in 'y'.
## i Row 2 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
## "many-to-many"' to silence this warning.
```

```
## 'summarise()' has grouped output by 'Test', 'Run_Date', 'Top_Std_Absorbance',
## 'Mean_PE_Conc'. You can override using the '.groups' argument.
```

```
Spike_Log_Plot <- ggplot(Joined_QAQC, aes(x = Run_Date, y = Spike_Percent_Diff)) +
  geom_point(size = 3, aes(col = Run)) +
  # geom_path(linewidth = 1) +
  facet_grid(~Test, scales = "free") +
  labs(title = "Spike") +
  theme_classic()
```

Spike_Log_Plot

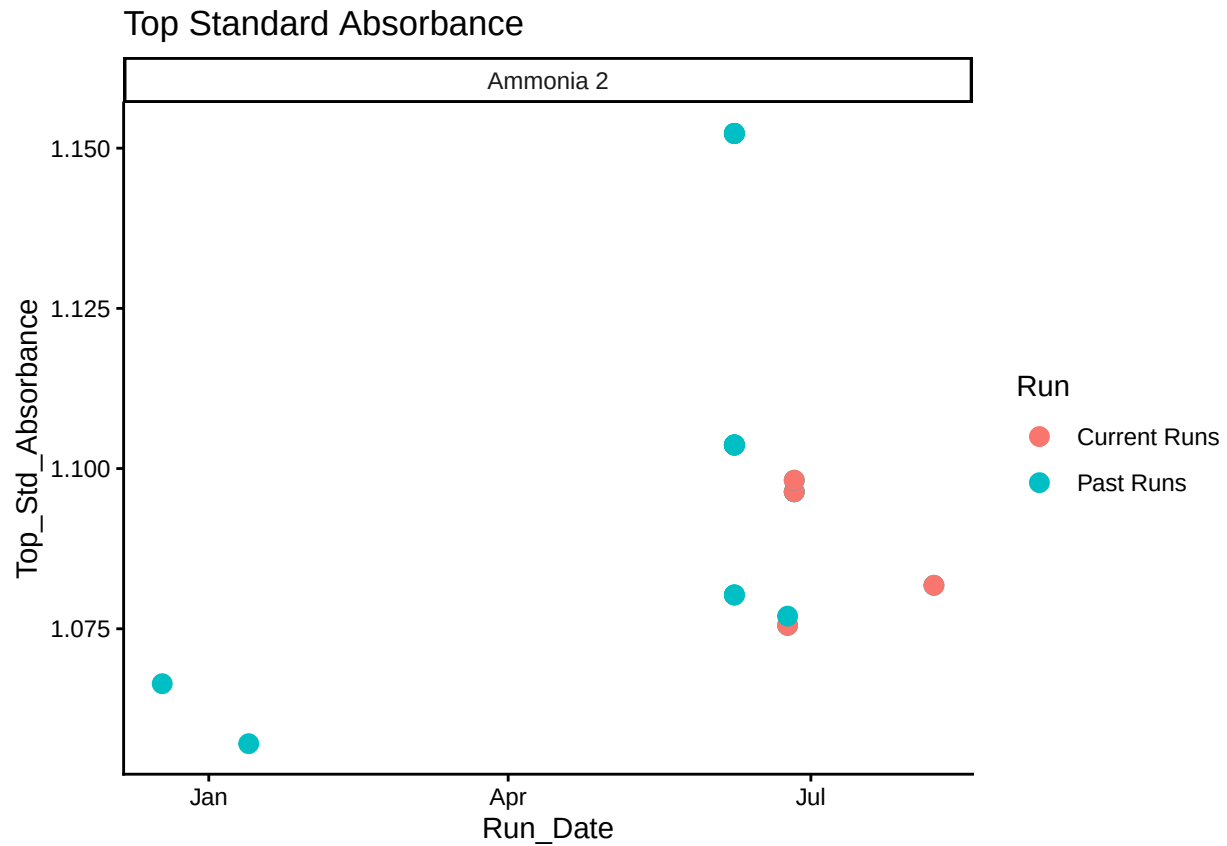
```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
TopStd_Log_Plot <- ggplot(Joined_QAQC, aes(x = Run_Date, y = Top_Std_Absorbance)) +
  geom_point(size = 3, aes(col = Run)) +
  # geom_path(linewidth = 1) +
```

```
facet_grid(~Test, scales = "free") +
labs(title = "Top Standard Absorbance") +
theme_classic()
```

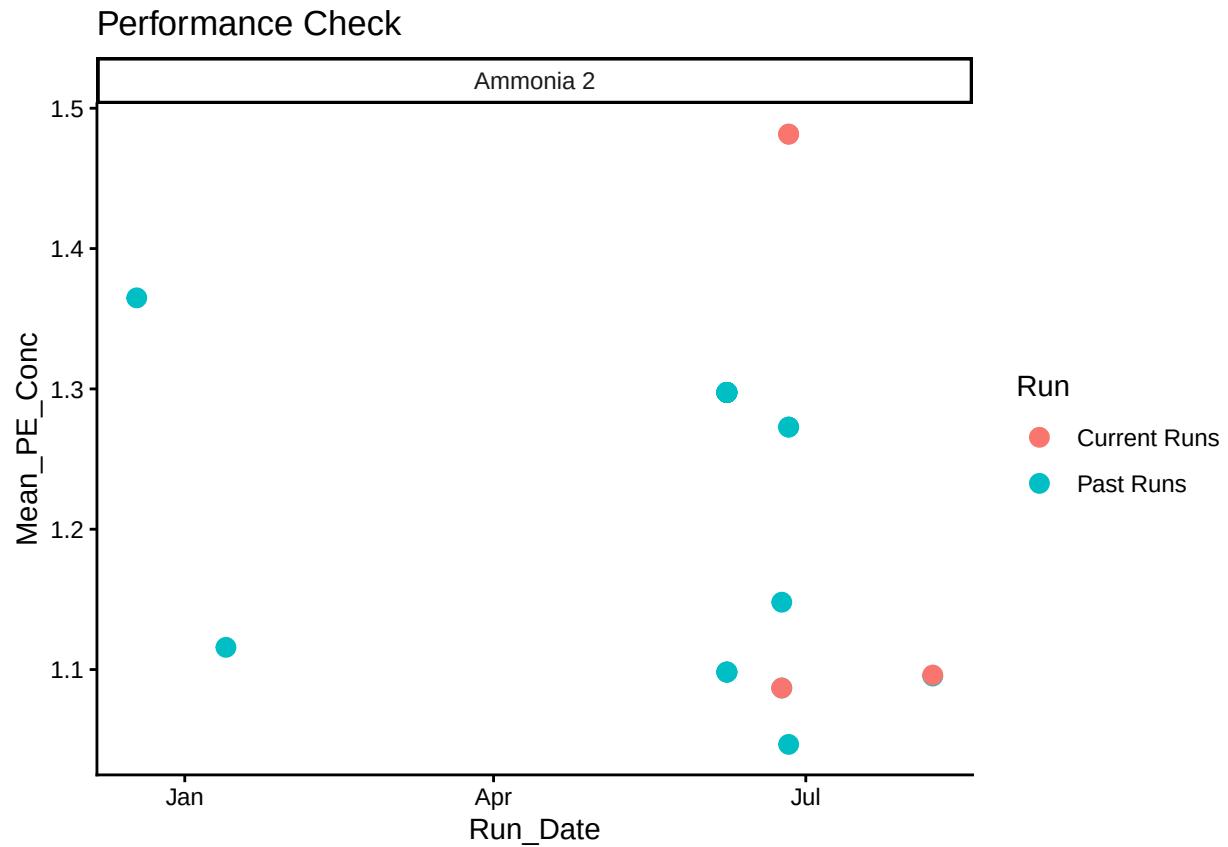
TopStd_Log_Plot



```
PE_Log_Plot <- ggplot(Joined_QAQC, aes(x = Run_Date, y = Mean_PE_Conc)) +
  geom_point(size = 3, aes(col = Run)) +
  #geom_path(linewidth = 1) +
  facet_grid(~Test, scales = "free") +
  labs(title = "Performance Check") +
  theme_classic()
```

PE_Log_Plot

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
##Write QAQC Log
```

0.11 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

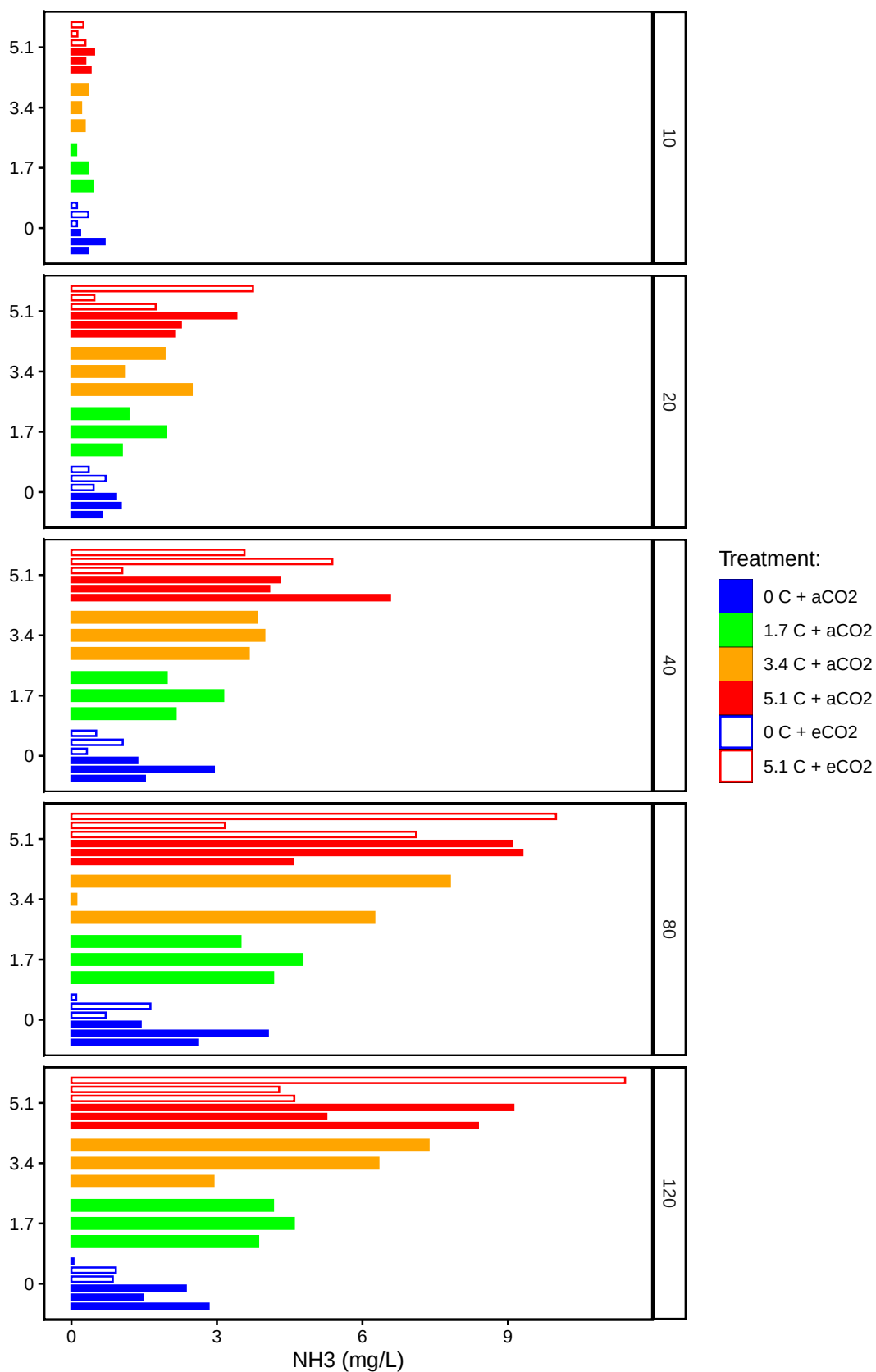
```
## Some sample IDs are missing from data.
```

```
## [1] "SMARTX_441-80"
```

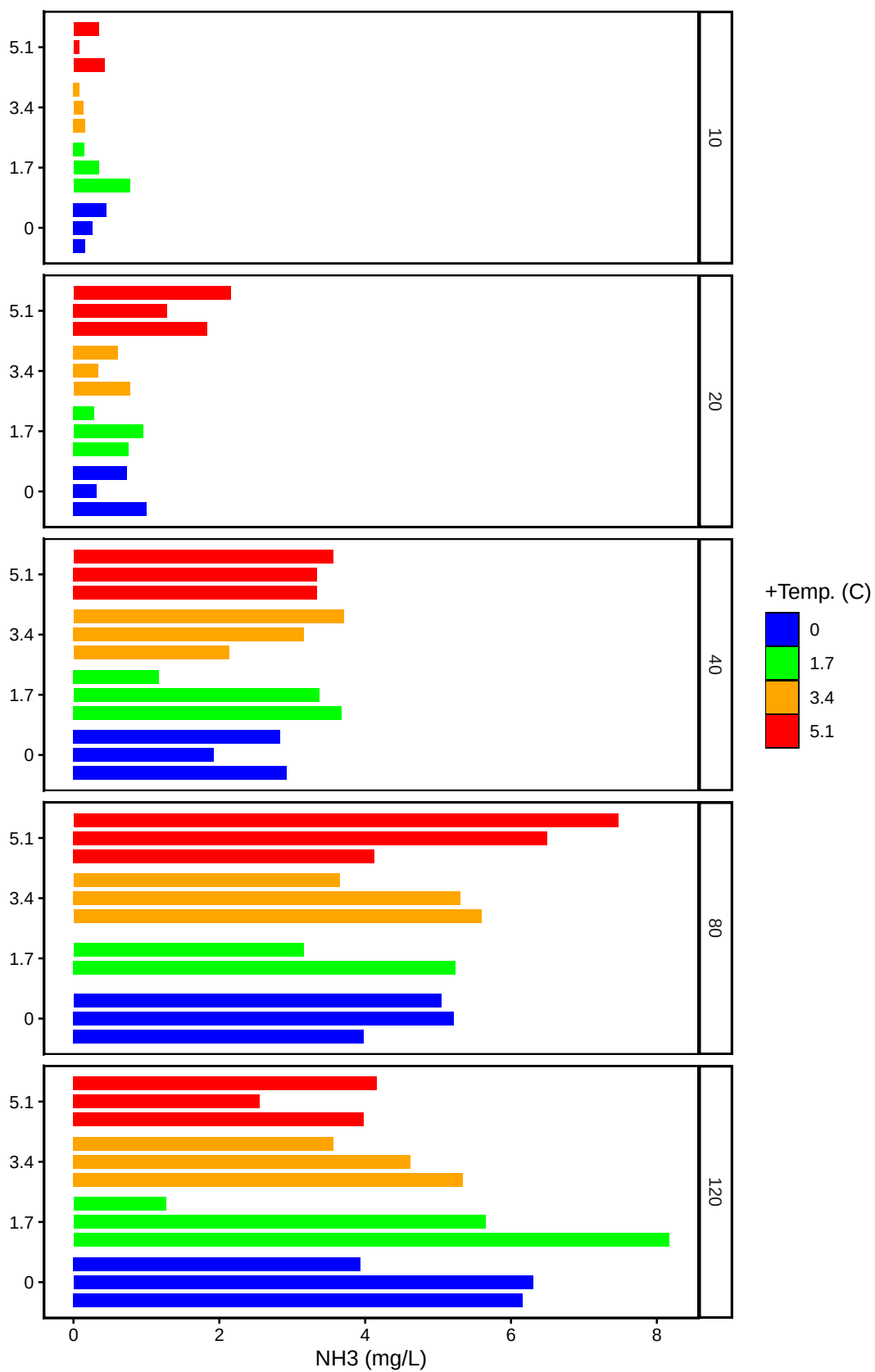
0.12 Visualize Data

```
## Visualize Data
```

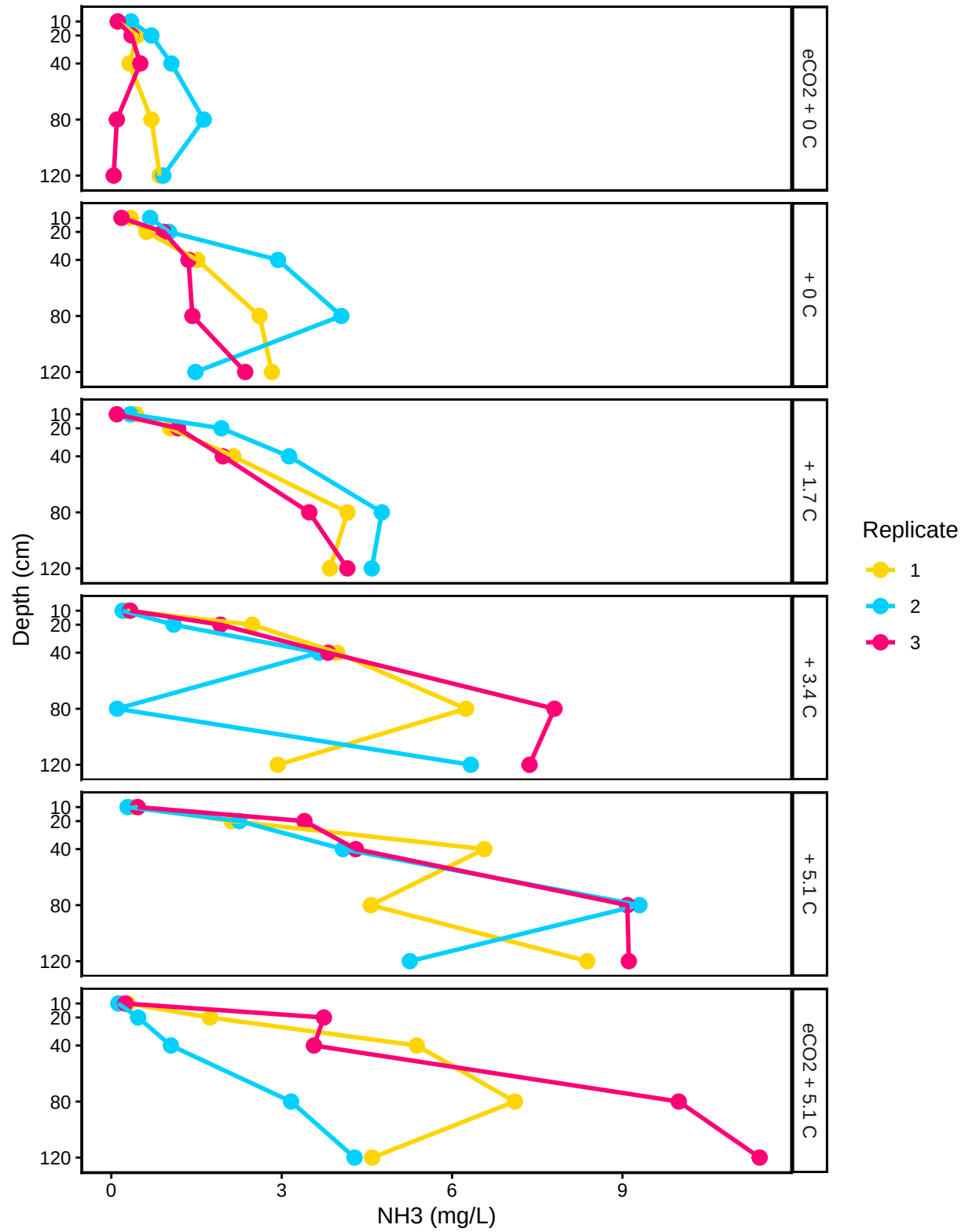

C3: Porewater NH3

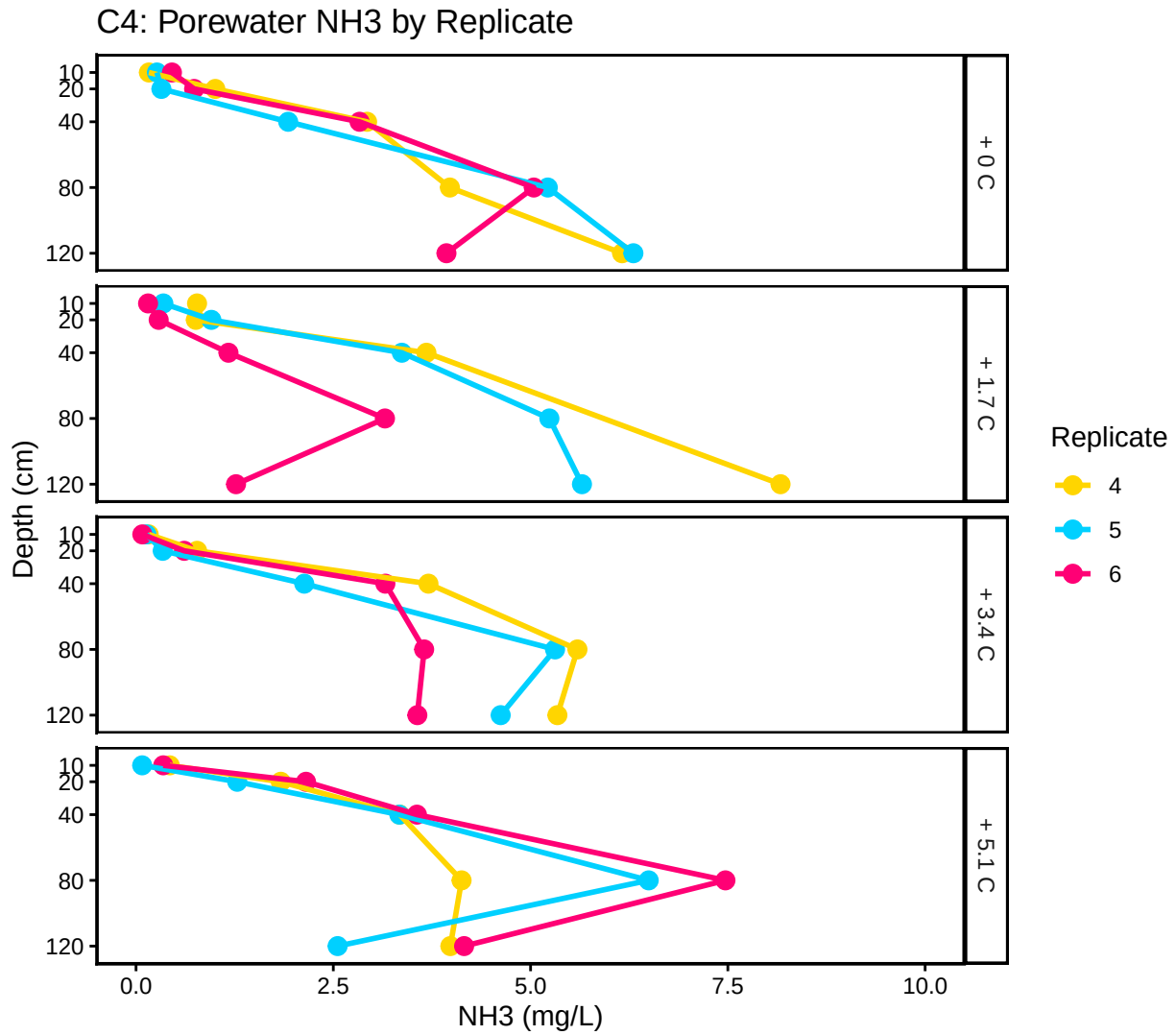


C4: Porewater NH3



C3: Porewater NH3 by Replicate





0.13 Organize

#Export Data

#end